

RCIM Model-Bank Reproduction Bidirectional Original-Dataset Mega Campaign Results

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

- campaign name: `track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17`
- planning report: `doc/reports/campaign_plans/track_1/exact_paper/2026-04-26-00-43-19_track1_bidirectional_original_dataset_mega_relaunch_after_micro_gate_plan_report.md`
- campaign output directory: `output/training_campaigns/track1/exact_paper/bidirectional_original_dataset/track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17`
- queue size: `400`
- launch mode: `remote_operator_launcher`
- remote host alias: `xilab-remote`

Completion Summary

- the bidirectional original-dataset exact-paper mega campaign completed the full `400/400` queue;
- the closeout refreshed both directional repository-owned restart matrices in `RCIM Paper Reference Benchmark.md` ;
- the closeout rebuilt the canonical paper-reference archives under `models/paper_reference/rcim_track1/forward/` and `models/paper_reference/rcim_track1/backward/` ;
- the closeout materialized campaign-level `campaign_leaderboard.yaml` , `campaign_best_run.yaml` , and `campaign_best_run.md` bookkeeping artifacts.

Campaign Best Run

- run instance id: 2026-04-26-17-34-18_track1_original_dataset_backward_ert_attempt_20_campaign_validation
- run name: track1_original_dataset_backward_ert_attempt_20
- direction label: backward
- winning family: ERT
- winning estimator: ExtraTreesRegressor
- mean component MAE: 0.033975
- mean component RMSE: 0.087169
- mean component MAPE: 14.8114%

Selection rule:

- Primary metric: winning_mean_component_mae_asc
- First tie-breaker: winning_mean_component_rmse_asc
- Second tie-breaker: winning_mean_component_mape_percent_asc
- Third tie-breaker: run_name

Benchmark Restart Surface

Table	Direction	Green	Yellow	Red	Total
Table 2 - Amplitude MAE	forward	82	2	16	100
Table 2 - Amplitude MAE	backward	76	7	17	100
Table 3 - Amplitude RMSE	forward	82	6	12	100
Table 3 - Amplitude RMSE	backward	81	6	13	100
Table 4 - Phase MAE	forward	82	0	8	90
Table 4 - Phase MAE	backward	61	5	24	90
Table 5 - Phase RMSE	forward	81	6	3	90
Table 5 - Phase RMSE	backward	63	5	22	90

Reference Archive Refresh

Direction	Family	Archived Targets	Unique Source Runs	Unique Source Configs	Archive Root
forward	SVM	19	11	11	models/paper_reference/rcim_track1/forward/svm_reference_models
forward	MLP	19	13	13	models/paper_reference/rcim_track1/forward/mlp_reference_models
forward	RF	19	12	12	models/paper_reference/rcim_track1/forward/rf_reference_models
forward	DT	19	12	12	models/paper_reference/rcim_track1/forward/dt_reference_models
forward	ET	19	12	12	models/paper_reference/rcim_track1/forward/et_reference_models
forward	ERT	19	10	10	models/paper_reference/rcim_track1/forward/ert_reference_models
forward	GBM	19	8	8	models/paper_reference/rcim_track1/forward/gbm_reference_models
forward	HGBM	19	11	11	models/paper_reference/rcim_track1/forward/hgbm_reference_models
forward	XGBM	19	11	11	models/paper_reference/rcim_track1/forward/xgbm_reference_models

Direction	Family	Archived Targets	Unique Source Runs	Unique Source Configs	Archive Root
forward	LGBM	19	13	13	models/paper_reference/rcim_track1/forward/lgbm_reference_models
backward	SVM	19	14	14	models/paper_reference/rcim_track1/backward/svm_reference_models
backward	MLP	19	10	10	models/paper_reference/rcim_track1/backward/mlp_reference_models
backward	RF	19	12	12	models/paper_reference/rcim_track1/backward/rf_reference_models
backward	DT	19	9	9	models/paper_reference/rcim_track1/backward/dt_reference_models
backward	ET	19	12	12	models/paper_reference/rcim_track1/backward/et_reference_models
backward	ERT	19	9	9	models/paper_reference/rcim_track1/backward/ert_reference_models
backward	GBM	19	9	9	models/paper_reference/rcim_track1/backward/gbm_reference_models
backward	HGBM	19	10	10	models/paper_reference/rcim_track1/backward/hgbm_reference_models
backward	XGBM	19	12	12	models/paper_reference/rcim_track1/backward/xgbm_reference_models
backward	LGBM	19	9	9	models/paper_reference/rcim_track1/backward/lgbm_reference_models

Linked Artifacts

- benchmark report: `doc/reports/analysis/rcim_paper_reference/RCIM Paper Reference Benchmark.md`
- training results master summary: `doc/reports/analysis/Training Results Master Summary.md`
- track1 reference root: `models/paper_reference/rcim_track1`
- campaign leaderboard:
`output/training_campaigns/track1/exact_paper/bidirectional_original_dataset/track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17/campaign_leaderboard.yaml`
- campaign best run YAML:
`output/training_campaigns/track1/exact_paper/bidirectional_original_dataset/track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17/campaign_best_run.yaml`
- campaign best run Markdown:
`output/training_campaigns/track1/exact_paper/bidirectional_original_dataset/track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17/campaign_best_run.md`
- active campaign state: `doc/running/active_training_campaign.yaml`

Closeout Notes

- the archive refresh rule remains mandatory for every future RCIM Model-Bank Reproduction closeout;
 - when a newly accepted target winner improves the stored archive entry under `models/paper_reference/rcim_track1/`, the archive must be updated together with its reference documents;
 - when the accepted target winner does not improve, the stored archive entry must remain unchanged.
- This report is the canonical closeout record for `track1_bidirectional_original_dataset_mega_campaign_2026-04-26_00_49_17` and is linked from `doc/reports/campaign_results/track_1/exact_paper/2026-04-27-12-06-34_track1_bidirectional_original_dataset_mega_campaign_results_report.md`.